

1. Administrative & Study Identification

Full Study Title: An Open-label, Multi-center, Phase 1/2 Study to Assess Safety, Efficacy and Cellular Kinetics of YTB323, a Rapid Manufacturing CAR-T Cell Therapy Targeting CD19 on B Cells, for Severe Refractory Systemic Lupus Erythematosus **Short Title/ Acronym:** CYTB323G12101 / YTB323 srSLE Study **Protocol Number:** CYTB323G12101 (EudraCT: 2022-001796-14) **NCT Number:** NCT05798117 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** YTB323 (rapcabtagene autoleucel), a rapidly manufactured CD19-directed CAR-T cell therapy **Phase of Development:** Phase 1/2 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safety, efficacy, and cellular kinetics of YTB323 in participants with severe refractory systemic lupus erythematosus (srSLE). **Secondary Objectives:** To evaluate preliminary efficacy utilizing disease activity scores (SLEDAI-2K, PhGA) and renal outcome measures, and to assess pharmacodynamics, including deep B-cell depletion, T-cell recovery, and relevant biomarkers (IgG, IgA, IgM, C3, C4, and anti-dsDNA levels).

2.2 Background & Rationale Systemic lupus erythematosus (SLE) is characterized by the production of autoantibodies against multiple self-antigens by autoreactive B cells. Patients with severe refractory SLE (srSLE) exhibit failure to respond to standard treatments, progressive organ damage, and high mortality. Traditionally manufactured CD19-directed CAR T-cell therapies have demonstrated the potential to promote full clinical remission in srSLE. YTB323 is a novel, rapidly manufactured autologous CD19-directed CAR T-cell therapy expected to provide improved safety, deep B-cell depletion, and robust disease control.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 24 evaluable patients (including an initial sentinel cohort of \$n=3\$). **Number of Sites:** Multi-center global study **Estimated Study Start:** April 2023 **Estimated Primary Completion:** February 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men and women aged ≥ 18 years and ≤ 65 years at screening, fulfilling the 2019 EULAR/ACR classification criteria for SLE. 2. Active (severe) disease defined by SLEDAI-2K ≥ 8 with at least one significant SLE-related organ involvement (e.g., renal, moderate or severe peri/myocarditis, pleuritis, or vasculitis). 3. Failure to respond to two or more standard immunosuppressive therapies (including mycophenolate or cyclophosphamide) and at least one biological agent. 4. Positive for at least one autoantibody: ANA at a titer of $\geq 1:80$, anti-dsDNA, or anti-Smith.

4.2 Exclusion Criteria 1. Clinically significant active, opportunistic, chronic, or recurrent infection within one month prior to or during screening. 2. Uncontrolled diabetes mellitus, lung diseases, or other illnesses jeopardizing the ability to tolerate lymphodepletion and CAR-T cell therapy. 3. Prior history of malignancy except for adequately treated localized basal cell/squamous skin cancer. 4. Pregnant or nursing females, or women of childbearing potential not using a highly effective method of contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single administered dose of 12.5×10^6 CAR-positive viable T cells. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single infusion with short-term acute observation and long-term follow-up (until CAR-T cells are no longer detectable for 12+ months).

5.2 Concomitant & Prohibited Therapy Permitted: Medications for CAR-T toxicity management (e.g., tocilizumab for Cytokine Release Syndrome) and supportive standard of care. **Prohibited:** Protocol-prohibited immunosuppressive or systemic therapies that could compromise CAR-T cell expansion and persistence.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Autoantibody labs, Leukapheresis
Baseline	Day -5 to -3	Lymphodepletion conditioning (Fludarabine/Cyclophosphamide)	Treatment
Interim	Day 0	Single IV Infusion of YTB323 (rapcabtagene autoleucel)	End of Study
End of Study	Month 12+	Long-term follow-up, disease activity scoring (SLEDAI), B-cell recovery evaluation	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability as measured by adverse events (AEs), severe AEs, deaths, vital signs, electrocardiogram, and laboratory assessments. **Measurement Method:** Clinical observation, grading of CRS/ICANS by standard criteria, and safety laboratory analysis.

7.2 Secondary & Exploratory Endpoints Cellular kinetics (expansion/Tmax and persistence of CAR-T cells measured by qPCR and flow cytometry). Preliminary efficacy assessed using disease activity scores (SLEDAI-2K), Physician's Global Assessment (PhGA), and renal outcome measures. Changes in biomarker levels including T cells, B-cell depletion/recovery, IgG, IgA, IgM, C3, C4, and anti-dsDNA.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive inpatient and outpatient monitoring for Cytokine Release Syndrome (CRS), Immune effector Cell-Associated Neurotoxicity Syndrome (ICANS), hypogammaglobulinemia, and lymphodepletion-related cytopenias (e.g., neutropenia, anemia). **Serious Adverse Events (SAEs):** Required to be reported immediately (typically within 24 hours). **Data Monitoring Committee (DMC):** Independent safety review committee to evaluate sentinel cohort data before broader expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive the YTB323 infusion). Efficacy evaluations will be performed on the treated population. **Sample Size Justification:** Exploratory Phase 1/2 trial (~ 24 target enrollment); powered to generate preliminary safety, cellular kinetics, and proof-of-concept efficacy rather than formal hypothesis testing. **Handling of Missing Data:** Missing data will be handled via standard imputation guidelines; emphasis is placed on descriptive statistics for safety and individual patient longitudinal plots for cellular kinetics.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring to ensure compliance with CAR-T specialized administration protocols. **Audit**

Trail: eCRF locking, source data verification, and GMP compliance tracking for autologous cell manufacturing and chain of identity.

11. Study Identifiers & Governance

Primary ID: CYTB323G12101 **Registry Number:** NCT05798117 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** CYTB323G12101 **Official Regulatory Title:** An Open-label, Multi-center, Phase 1/2 Study to Assess Safety, Efficacy and Cellular Kinetics of YTB323, a Rapid Manufacturing CAR-T Cell Therapy Targeting CD19 on B Cells, for Severe Refractory Systemic Lupus Erythematosus

12. Clinical Framework (SLE Specific)

Primary Condition: Severe Refractory Systemic Lupus Erythematosus (srSLE) **Diagnosis Threshold:** SLEDAI-2K ≥ 8 and failure of at least 2 immunosuppressants + 1 biologic **Medical Methodology:** Rapid Manufacturing Autologous CD19-directed CAR-T Cell Therapy **Optimization Target:** Deep B-cell depletion, resolution of autoantibodies, and full clinical remission

13. Study Procedures & Methodology

Management Pathway: Apheresis $\rightarrow$ Rapid manufacturing of YTB323 $\rightarrow$ Lymphodepletion $\rightarrow$ CAR-T Infusion **Education Protocol (HCP):** Specialized site training on CAR-T administration, CRS/ICANS recognition, and rescue agent (tocilizumab) deployment. **Education Protocol (Patient):** Counseling on long-term cytopenias, infection risks, contraception requirements, and the 15-year gene therapy long-term follow-up mandate. **Quality Audit Mechanism:** GMP release criteria tracking for cell viability, and stringent chain-of-custody oversight for autologous cells.

14. Observation & Follow-Up Schedule

Total Duration: Primary follow-up up to 12-24 months; extended long-term follow-up as required for gene therapy. **Onsite Visit Cadence:** Intensive monitoring Days 0-28; subsequent visits at Months 3, 6, 9, 12. **Remote Monitoring Frequency:** Periodic telehealth check-ins or local laboratory draws (qPCR tracking). **Checklist Review Interval:** Routine tracking of infection parameters, cytopenias, and SLEDAI-2K updates during follow-up.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					